

# Technical guide (step by step, with the objects and libraries involved)

Author: Pankaj Yadav, Forward Deployed Software Engineer · [github.com/Pankkaj64](https://github.com/Pankkaj64) · [linkedin.com/in/pankkaj64](https://linkedin.com/in/pankkaj64) · resume: docs/pdf/Pankaj-Yadav-Resume.pdf

Companion to docs/pdf/how-it-works.pdf . This document follows the same flow but names the files, functions, objects and libraries at each step. Line numbers are approximate; function names are exact.

## Stack

| Layer                  | Library                                                                                 | Why                                                                  |
|------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Web UI                 | React 19, Vite 6, TypeScript 5.7                                                        | one page, hot reload, typed packet                                   |
| Speech                 | browser <code>SpeechRecognition</code> , <code>speechSynthesis</code>                   | no audio leaves the browser; no key needed                           |
| API                    | FastAPI, uvicorn, pydantic 2                                                            | validated request bodies, Server-Sent Events                         |
| HTTP client            | httpx (async)                                                                           | Ollama, eBay, Google Books, Open Library, FX                         |
| Images                 | Pillow                                                                                  | crops, rotations, JPEG encoding, quality heuristics                  |
| Detection              | ultralytics (YOLO26s, YOLO11s, YOLOE-26s) on CPU                                        | book and object boxes                                                |
| OCR                    | PaddleOCR (PP-OCRv6), EasyOCR as alternative                                            | spine text                                                           |
| Vision and chat models | Ollama: <code>gemma3:latest</code> , <code>qwen3-vl:2b</code> , <code>qwen2.5:3b</code> | title proposal, blind crop check, conversation                       |
| Config                 | python-dotenv                                                                           | <code>.env</code> first, <code>config.py</code> defaults as fallback |
| Formatting             | Ruff                                                                                    | import order and formatting ( <code>ruff</code> check app tools )    |

## Repository map

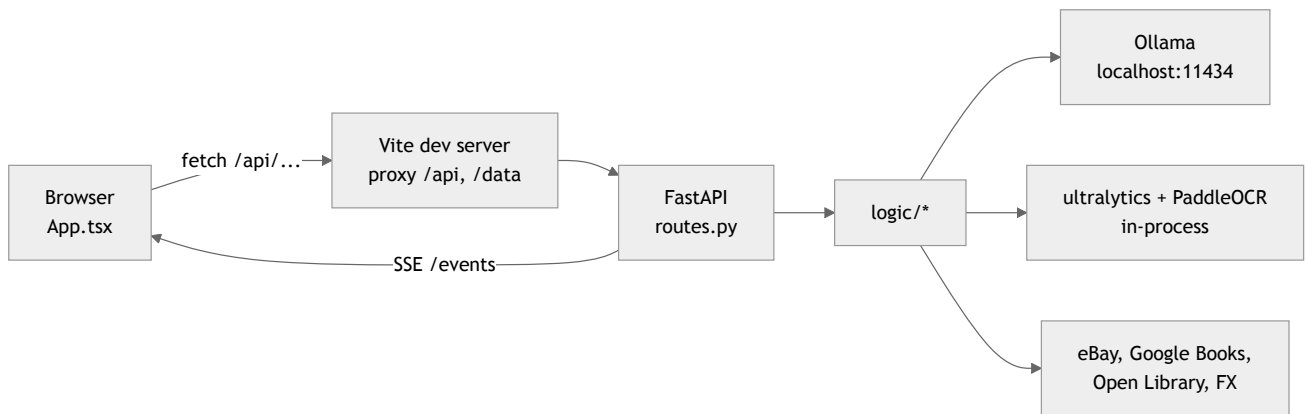
|                                      |                                                            |
|--------------------------------------|------------------------------------------------------------|
| backend/app/config.py<br>(env first) | paths, model names, room classes, API endpoints, Settings  |
| backend/app/main.py                  | FastAPI app, CORS, request debug prints, health, startup   |
| warm-up, static mounts               |                                                            |
| backend/app/routes.py                | every HTTP route (transport only)                          |
| backend/app/logic/<br>state.py       | SWEEPS store, checkpoints, start/finish, background        |
| research                             |                                                            |
| perception.py                        | quality, detection, OCR per crop, title proposal, count    |
| check, inspect_frame                 |                                                            |
| ocr.py                               | PaddleOCR / EasyOCR behind read_text()                     |
| crop_verifier.py                     | blind category check via Ollama                            |
| identification.py                    | identity gate rules, Open Library lookups                  |
| tracking.py                          | appearance signatures, cross-frame matching                |
| workflow.py                          | merge_observations, refresh_workflow (measurement ->       |
| pricing -> validation), guidance     |                                                            |
| conversation.py                      | Turn model, command matching, voice tools, chat model call |
| measurement.py                       | reference-line and shelf-width scale, room geometry        |
| pricing.py                           | Quote/Offer models, matching, locale table, apply_prices   |
| providers.py                         | eBay, Google Books, FX, research, second-country           |
| comparison                           |                                                            |
| claim.py                             | SweepStart, empty_packet, totals, validate_packet,         |
| build_claim_packet                   |                                                            |
| report.py                            | report_html, claim_bundle                                  |
| evaluation.py                        | scoring against ground truth                               |
| utils.py                             | stamp(), trace_step, pipeline_stage, performance_summary   |
| frontend/src/                        |                                                            |
| App.tsx                              | the page: start/stop, capture loop, SSE, voice wiring      |
| api.ts                               | get/post helpers, API base URL                             |
| voice.ts                             | speak(), listen(), barge-in                                |
| types.ts                             | Packet, Book, Item, Locale types                           |
| components/Inventory.tsx             | counts and line list                                       |
| components/Conversation.tsx          | transcript and text fallback                               |

## The central object: `packet`

One Python `dict` per sweep, created by `empty_packet()` in `claim.py`, stored as `SWEEPS[sweep_id]["packet"]` in `state.py`. Every stage mutates it in place. The first six keys are the output contract (`sweep`, `room`, `books`, `items`, `totals`, `review_queue`); the rest is working state (`frames`, `transcript`, `calibrations`, `guidance`, `price_details`, `research`, `quotes`, `fx_rates`, `audit_trail`, `stage_runs`, `performance`, ...).

It is checkpointed to `data/packets/<id>.active.json` after every change, streamed to the browser over SSE, and finally filtered by `build_claim_packet()` into `claim_packet.json`.

## Request flow



In development the browser only talks to Vite; `vite.config.ts` forwards `/api` and `/data` to `127.0.0.1:8000`, so no CORS is needed. In a production build FastAPI serves `frontend/dist` itself.

## Step 1: configuration (`config.py`)

- `load_dotenv(BACKEND_DIR / ".env")` then `model_choices.env`. Shell and `.env` values win.
- Module constants `DEFAULT_*` are the fallbacks; Settings properties read `os.getenv(name, DEFAULT)` lazily, so a value is resolved on first use.
- `ROOM_CLASSES` is the prompt list fused into the YOLOE weights by `tools/setup_detector.py`.
- Endpoint constants (`GOOGLE_BOOKS_VOLUMES_URL`, `EBAY_API_HOSTS`, `OPEN_LIBRARY_*`, `FRANKFURTER_URL`, `OPEN_ER_API_URL`) are overridable by environment variables of the same name.

## Step 2: startup (`main.py`)

1. `FastAPI(...)`, `CORSMiddleware` from `settings.cors_origins`.
2. `log_request` middleware makes a short `request_id` and prints `http.received`, `http.responded` (with status and elapsed time) or `http.failed` for every request.
3. `@app.on_event("startup")` schedules `perception.warm_models()`: loads the OCR engine in a thread and asks Ollama to load the three models with `keep_alive: 30m`, so the first frame is not a cold start.
4. Routers are included before the static mounts (`/data` for evidence, `/` for the built UI), otherwise the catch-all mount would swallow API paths.
5. `state.restore_active_sweeps()` reloads unfinished sweeps from their checkpoints.

## Step 3: create a sweep

Browser (`App.tsx`, `start()`):

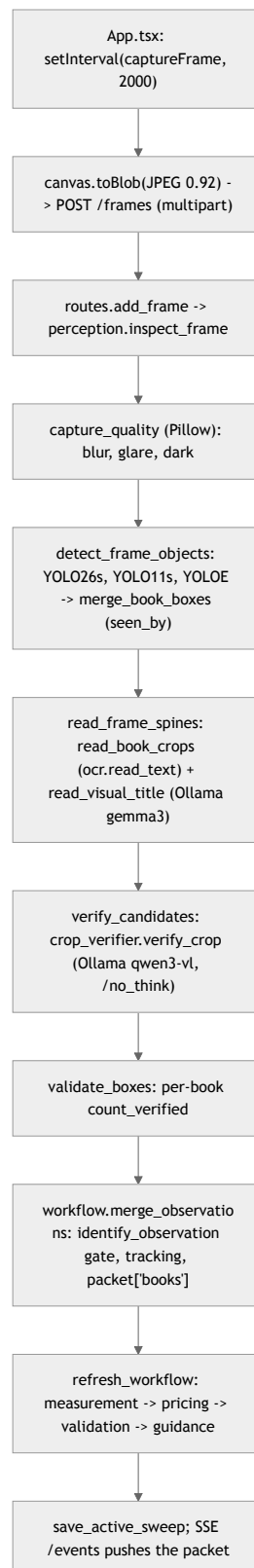
```
navigator.mediaDevices.getUserMedia({ video: { facingMode: "environment", width: {  
ideal: 1920 } } })  
post("/api/sweeps", { country, currency, country_code, appraisal_threshold, device })  
setSweepId(id); setPacket(packet); setRunning(true)  
speak(greeting); listen(sendTurn, stopSpeaking)
```

Server ( routes.create\_sweep -> state.start\_sweep ):

1. Body validated as `SweepStart` (pydantic: threshold > 0, country code `^[A-Z]{2}$` or empty).
2. `uuid.uuid4()` becomes the sweep id; `empty_packet()` builds the contract skeleton.
3. `SWEEPS[id] = {"packet", "started", "position", "status": "active"}`.
4. `refresh_workflow(packet)` runs validation once so the first packet already has findings.
5. `save_active_sweep(id)` writes the checkpoint atomically (temp file + rename).

## Step 4: the frame loop

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Details per stage:

- **Capture (App.tsx):** `busyRef` allows one upload in flight; the interval is 2 s but the real cadence is "when the previous frame returns". The shelf label from the input box is sent as a form field.
- **Route (routes.add\_frame):** saves the JPEG to `data/frames/<sweep>-<hash>.jpg`, calls `inspect_frame(raw, ref)`, merges the result into the packet, returns `{packet}`.

- `inspect_frame (perception.py)`: runs under `_detect_lock` (one frame at a time on an 8 GB laptop) and wraps each stage in `pipeline_stage(candidate, name)`, which records elapsed seconds and errors into `frame["pipeline_stages"]`.
- `detect_objects(raw, model_name)`: lazy from `ultralytics` import `YOLO`, `YOLOE`; weights from `backend/.runtime/models`; `predict(imgsz=960, conf=0.15, device="cpu")`; returns normalised `bbox + category + confidence`. Models are cached in `_detectors`.
- `select_objects`: splits into books (de-duplicated by `IoU > 0.65`, `partial` flag at the image edge) and items (non-book, not in `IGNORED_CLASSES`).
- `merge_book_boxes(books, found, detector)`: a box from another detector either joins as a new candidate or adds its detector name to an existing box's `seen_by` list.
- `read_book_crops`: crop with 4 px padding, `thumbnail((640, 640))`, OCR at 0/90/270 degrees, stop early when a line has confidence  $\geq 0.9$ , keep the orientation with the highest confident-character score, bounded by `OCR_FRAME_BUDGET_S = 15`.
- `ocr.read_text(bytes)`: one engine per process behind a lock; PaddleOCR 3.x `predict()` results read from `rec_texts / rec_scores`; lines under confidence 0.5 dropped. `USE_TF=0` is set before import to avoid a TensorFlow/Keras 3 conflict.
- `read_visual_title(raw, book)`: crop to 672 px, JSON schema `{is_book, title, author, publisher}` where author/publisher are an enum of the OCR lines, `think: false`, 90 s timeout, `keep_alive: 30m`; results cached by appearance signature in `_reader_cache`.
- `verify_crop(crop, detector_category, model)`: prompt without the label, schema `{category, clear_single_object, visible_features}`, `verification_decision` requires the category to match and named parts (cover, spine, text) in `visible_features`; cached by SHA-256 of the crop. Frame budget `CROP_VERIFICATION_FRAME_BUDGET_S`; the rest is deferred.
- `validate_boxes(raw, primary, all_books)`: runs the validator detector again, `compare_boxes` for the frame-wide note, and sets `book["count_verified"]` when the box is matched by the validator (`IoU >= 0.5`) or `seen_by` has two detectors.
- `identify_observation (identification.py)`: `status = "identified"` only if `title.casefold() in ocr_text`, `confidence >= 0.75`, `count_verified` (fallback: frame agrees) and `identity_verified` (reader title == OCR-supported title, crop check agreed).
- `reading_agreement` has two evidence-bound paths, recorded on the line as `identity_basis`: `ocr_exact` (the proposal is literally in the OCR text at  $\geq 0.9$ ) and `two_models` (the reader's title equals the blind check's `visible_text` and `ocr_coverage(title, lines) >= 0.3` with one matching run of four letters or more). Both require `is_book`,  $\geq 8$  characters, a whole spine (`partial` false) and a title that is not the author line. `misread_word` refuses a title word the blind check read differently (edit ratio  $\geq 0.8$ ), writing `title_rejected`.
- **Cross-frame pooling** (`tracking.merge_book_evidence`): when `match_track` associates a frame book with an existing line (appearance signature, title similarity, or the scene transform projecting the old box onto the new frame with overlap  $\geq 0.45$ ), the line keeps `partial = old` and `new`, the union of `seen_by`, pooled `ocr_lines` (best confidence per text), the first title proposal and the strongest blind check; `merge_observations` then calls `promote_verified_book` on the pooled line.

- **Capture guidance** (`workflow.next_guidance`): codes `edges` (a third or more of the frame's boxes are `partial`) and `closer` (median box height under 12 % of the frame with eight or more boxes) come before `retake` and `unreadable`.
- **OCR** (`perception.prepare_spine`): crops under 160 px on a side or 700 px tall are upsampled two times (LANCZOS), capped at 1024 px, `ImageOps.autocontrast(cutoff=1)`; rotations 0, 90, 270 and finally 180 with the existing early stop.
- **OCR languages** (`ocr._load_engine(lang)`, `perception.read_rotations`): one engine per language (`OCR_LANG`, default `en`, plus `OCR_EXTRA_LANGS`, default `hi`, PaddleOCR's `devanagari_PP-0CRv5_mobile_rec`). The crop is read upright in English; if no line reaches 0.9 with four characters, each extra language is read upright next; the language that is confident gets the remaining rotations, otherwise the higher-scoring language wins. Each OCR line carries `lang`. `ocr_budget_s` multiplies the per-frame budget by the number of languages and has a 30 s floor. The startup warm-up loads every language.
- **Word matching** (`utils.normalized`): `\w` alone drops combining marks and split Hindi words at vowel signs ("श्रीमद्भगवद्गीता" became "श र भद ..."); the `WORD` pattern keeps the Indic blocks U+0900–U+0DFF and combining diacritics, so title, coverage, misread and catalogue checks work on Hindi text.
- **Tiles** (`perception.detect_tiles`): four overlapping tiles per frame for all three detectors, boxes mapped back, half boxes at tile borders dropped; behind `DETECTOR_TILES` (default off, see failure log 18).
- **Catalogue witness** (`identification.catalogue_witness`): at finish, Open Library search with the title and author; `confirmed` when every title word is in a catalogue title, `misread` when a title word is absent but within edit ratio 0.8 of a catalogue word, in which case `providers.demote_misread` clears the title and flags the line.
- **Author is never the title**, enforced three times. `validated_reading` (`perception.py`) empties the title when the reader's title equals its own author choice; `reading_agreement` refuses a title equal to the author line; and at finish `resolve_work` first asks the Open Library authors search (`OPEN_LIBRARY_AUTHORS_URL`): a title whose exact name is an author with 20+ works returns `author_as_title`, `demote_author_title` (`providers.py`) moves the text to `author`, sets `status = "unidentified"` and `title_rejected`, and `run_research` copies that to the live packet with a review finding. Finish runs research whenever `ENABLE_CATALOGUE_LOOKUP` is on, even with no price keys.
- `merge_observations` (`workflow.py`): matches frame books to existing inventory lines by appearance (`tracking.py`) and scene alignment, creates `book` dicts with contract fields plus evidence fields, appends review findings and notes.
- `refresh_workflow`: `stage(packet, "measurement") -> measure_spine` where a calibration exists; `stage("pricing") -> apply_prices`; `stage("validation") -> validate_packet`; then `next_guidance` sets `packet["guidance"]`.
- **Two pages** (`App.tsx`, view `state synced with` `location.hash`): the sweep page is a two-column grid (controls, camera, status strip and live counts left; the chat with the agent right) and the inventory page (`#inventory`: claim packet links and summary, then the full inventory) opens from the header button once `packet.sweep.finished_at` is set. The browser back button returns to the sweep; the sweep id is kept in `sessionStorage`.

- **Status strip and Stop** (`App.tsx`): `inFlight` while a frame upload awaits the backend, finishing from the Stop click until the packet arrives; `stop()` waits on `busyRef` for the in-flight frame, then calls `stop-capture` and `finish`, retrying a 409 "Wait for frame processing" every 2 s for up to five minutes.
- **Responsive layout** (`style.css`): wide desktop ( $\geq 1400$  px), desktop, tablet ( $\leq 960$  px, one column), phone ( $\leq 600$  px: stacked fields, 44 px controls, 16 px inputs, two-line inventory rows), small phone ( $\leq 380$  px), landscape phones, `pointer: coarse`, `prefers-reduced-motion`, notch safe areas with `viewport-fit=cover`, print stylesheet.
- **SSE** (`routes.workflow_events`): an async generator yields `data: <json>` whenever the serialised packet changes, with `retry: 1500` and a lifetime limit; `EventSource` in `App.tsx` applies it with `setPacket` and speaks new guidance when `!isSpeaking()`.
- **Providers** implement the `PriceProvider` protocol and return `Quote` objects (`source`, `url`, `retrieved_at`, `condition`, `currency`, `amount`, `match_score`, `basis`). eBay uses an OAuth client-credentials token cached in `EbayAuth`; the median of matching fixed-price listings is the quote and every comparable is kept. Google Books uses the two-letter `country` parameter; failures record HTTP `<status>` and a hint.
- **Catalogue ISBN** (`providers.resolve_catalogue_isbn`): run in `research_inventory` for an identified line with no visible ISBN, after the author-as-title and misread checks. Google Books (plain query, 20 results, 503 retry) then Open Library; a candidate fits when its normalised title equals or starts with the identified title and, if an author is known, contains the author's surname; English-language volumes (Google `language: en`, ISBN prefixes 9780/9781) are preferred; the ISBN must pass `identification.valid_isbn`. The result goes to `result["isbn_resolution"]` and the line's `catalogue_isbn`, `isbn_source` and `isbn_url`; `run_research` copies those three keys to the live packet. The contract's `isbn` is the visible ISBN or else the catalogue ISBN, with `isbn_source` saying which. `RESOLVE_CATALOGUE_ISBN=false` disables the step.
- **eBay search order** (`EbayProvider.get_book_quotes`): visible ISBN, catalogue ISBN, then title + author; the next search runs only when the previous one found fewer than two listings above `PRICING_MIN_MATCH`. Every search is recorded in `attempts[kind].searches` and each quote carries `search_basis`. `limit` is 50.
- **Range** (`providers.price_range`): count, low, p25, median, p75 (inclusive quartiles) and high of the matching listings in their own currency; `Quote.price_range` carries it, `pricing._priced` converts it with the selected quote's dated rate, and the contract adds `replacement_range / used_range`; `report.range_note` prints "N listings: low-high, middle half p25-p75".
- `apply_prices` is pure (no network, no model): it fills `replacement_cost` and `used_value` or the `EMPTY_*` shapes, and writes candidates, rejections and the chosen quote into `packet["price_details"][line_id]`.
- `validate_packet` appends rule findings to `review_queue` (unreadable, below threshold, no sourced price, missing URL/date, currency mismatch, converted without FX, implausible dimensions, duplicates, missing frame, unknown room scale, unconfirmed coverage).
- `calculate_totals` uses `Decimal` with `ROUND_HALF_UP`; unknown values are excluded and counted in `excluded_from_totals`.

- `build_claim_packet` keeps the required keys first and in order, trims lines to the contract fields ( `null` / `""` for unknowns), drops excluded lines, then appends `price_details`, `locale_comparison`, `evidence_frames`, `performance`, `cost`, `methodology`. `report_html` renders from this; `claim_bundle` zips packet, report, frames and a SHA-256 manifest.

## Why small local models, not bigger ones

The system is built and measured on an **Apple M3 laptop with 8 GB of memory**, and the brief expects a reviewer to run it from the README on their own laptop. On Apple silicon the CPU and GPU share that 8 GB, so every model competes with macOS, the browser and the backend.

| What is loaded                     | Memory                             |
|------------------------------------|------------------------------------|
| Gemma 3 4B (title reader)          | 3.3 GB                             |
| Qwen3-VL 2B (blind check)          | 1.9 GB on disk, 2.7 GB once loaded |
| Qwen2.5 3B (conversation)          | 1.9 GB                             |
| Three YOLO detectors               | about 70 MB                        |
| PaddleOCR, English and Hindi       | about 220 MB                       |
| macOS, the browser and the backend | the rest of the 8 GB               |

- **A bigger vision model does not fit.** Qwen2.5-VL 3B, the first choice, cannot load: Ollama's Metal backend asks for a 6.6 GB buffer. With a 1k context it loaded but took 57–160 s per crop. 7B and larger vision models need more memory than the machine has.
- **The small models already fill the memory.** Together they need about 7 GB, so macOS swaps when all three are active. That is why OCR took 107 s for one frame inside the app but 2–8 s per crop on its own, and why each model gets a per-frame time budget.
- **Hosted large models were rejected on purpose.** A cloud model (GPT-4o, Gemini, Claude) would read worn spines better, but every frame of the claimant's home would leave the device, each sweep would cost money per frame, and a network round trip would sit in the live loop. Running locally keeps model cost at zero and the video on the laptop.
- **What this costs.** Small models read fewer worn titles. The design compensates by refusing rather than guessing: a title needs two independent readings, so a weaker model lowers the identification rate but not its honesty.
- **Swapping is configuration only.** On a machine with 16 GB or more, or a GPU, larger models can be set with `BOOK_READER_MODEL`, `CROP_VERIFIER_MODEL` and `CONVERSATION_MODEL` in `backend/.env`; no code changes.

## Observability

- There is no logging framework. Every step prints one debugger-style line to stdout: `[DEBUG 14:07:11.851] backend.ready detector=yolo26s.pt recovered_sweeps=6`. Warnings use `[WARN ...]` and errors `[ERROR ...]`; `stamp()` in `utils.py` supplies the wall-clock time.

- `@trace_step("name")`: prints `.started`, `.finished` (elapsed), `.failed`, `.cancelled`, and a `.waiting` heartbeat every 10 s for async steps, with `sweep_id` / `frame_ref` when present.
- `pipeline_stage` (per frame) and `stage` (per workflow pass) write timings into the packet; `GET /{id}/performance` summarises calls, mean and max per stage, and `time_to_packet_s`.
- `GET /api/health` reports which models and price providers the backend will use.

## API summary

| Method and path                                                                                                               | Purpose                                                       |
|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <code>POST /api/sweeps</code>                                                                                                 | create a sweep ( <code>SweepStart</code> )                    |
| <code>GET /api/sweeps</code> , <code>GET /api/sweeps/locales</code>                                                           | saved sweeps; supported countries                             |
| <code>GET /api/sweeps/{id}</code> , <code>GET .../events</code>                                                               | packet; Server-Sent Events stream                             |
| <code>POST .../frames</code> , <code>.../video</code>                                                                         | one JPEG (multipart); optional full recording                 |
| <code>POST .../turns</code> , <code>.../corrections</code>                                                                    | one spoken or typed turn                                      |
| <code>POST .../calibration</code> , <code>.../spine-bounds</code> , <code>.../room</code> , <code>.../item-measurement</code> | explicit measurements                                         |
| <code>POST .../offers</code> , <code>.../research</code> , <code>GET .../prices</code> , <code>POST .../compare-locale</code> | pricing                                                       |
| <code>POST .../inventory-review</code> , <code>.../settings</code>                                                            | operator review and runtime settings                          |
| <code>POST .../stop-capture</code> , <code>.../finish</code>                                                                  | end capture; run background stages and export                 |
| <code>GET .../claim-packet</code> , <code>.../report</code> , <code>.../bundle</code> , <code>.../performance</code>          | deliverables and timings                                      |
| <code>POST .../evaluate</code>                                                                                                | score against a ground-truth JSON file (format in the README) |

## Running and testing

```
cd backend && pip install -r requirements.txt && python tools/setup_detector.py
ollama pull gemma3:latest && ollama pull qwen3-vl:2b && ollama pull qwen2.5:3b
uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
cd ../frontend && npm install && npm run dev
# formatting and lint
cd backend && ruff check app tools && ruff format app tools
```

The measured model choices and the known limits are summarised in the README.